

TITLE PAGE

- **Problem Statement ID** – 26089
- **Problem Statement Title**-Cooperative Gig Services Platform for Household & Community Services
- **Theme**-Agriculture, FoodTech & Rural Development
- **PS Category**- Software
- **Team ID**-
- **Team Name**- TransitIQ



PROPOSED SOLUTION-

Our Idea is an web based application that will connect verified Cooperative Workers to Customers.

- Book a verified worker in a minute- Electrician,Plumber,Maid,Caregiver all in one app,in one go.
- **Safety Concern**-live location of workers are traced ,ID'S are checked properly .
- **Fair Prices** always-no last minute price shock.
- Honest **Rates and Reviews**.
- Workers get real **income and insurance**
- Workers can manage their **profile,skills and service**.



Problem it addresses-

- **Cooperative workers are invisible online**-a dedicated app puts them in front.
- **Emergencies**(leaks,electrical faults)-have no quick fix.
- **Smart Matching** based on skills ,location and availability.
- **Transparent pricing**.
- **Better opportunities for workers**.
- **Reduced dependency on informal Networks**-customers don't have to depend only on personal reference to find workers.

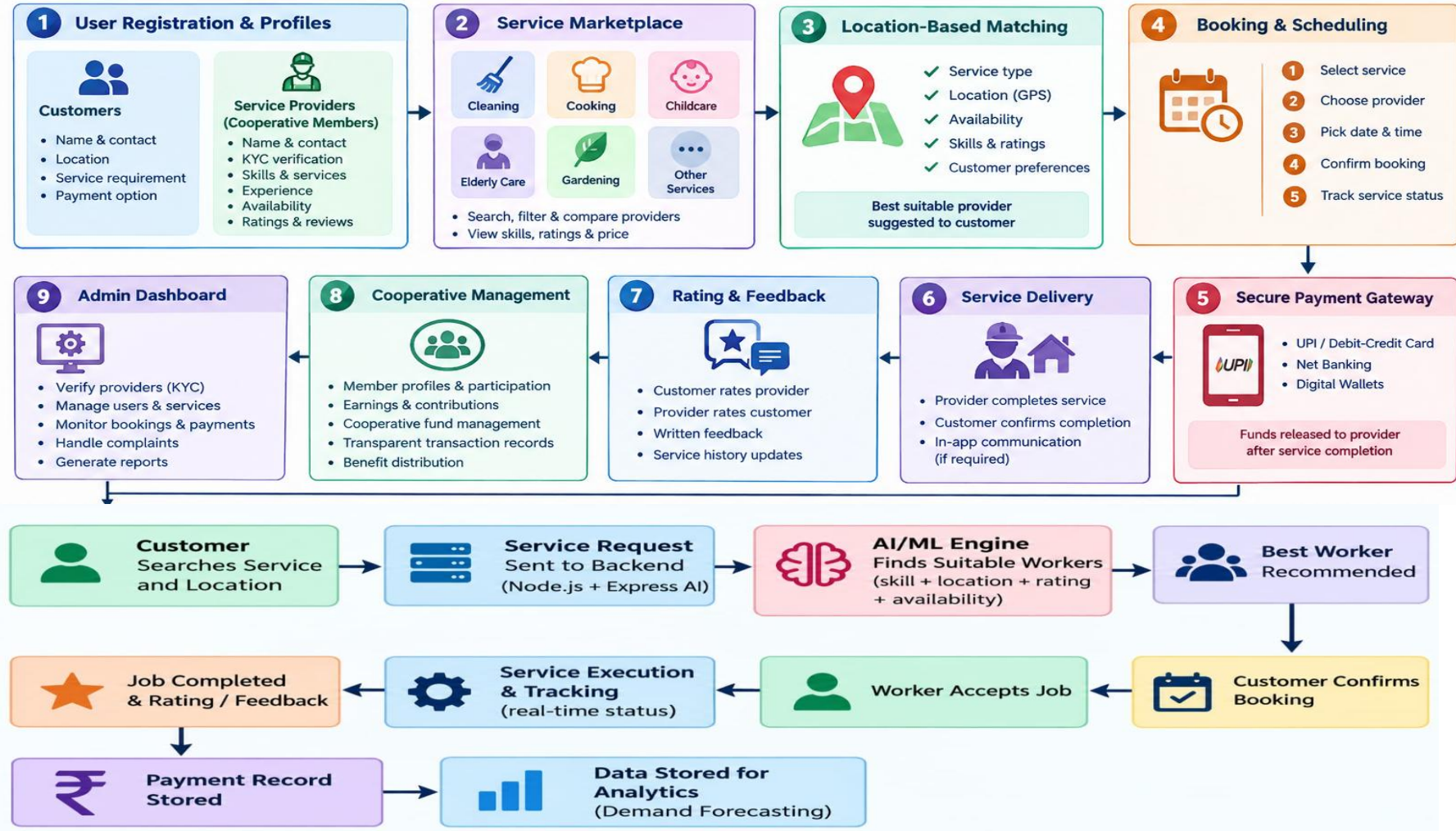
INNOVATIVE AND UNIQUENESS

- **Cooperative-first platform** specifically designed for labour cooperatives.
- **AI demand forecasting** –predict upcoming service demand and help cooperative plan their workforce.
- Combines **booking +worker management +tracking+reviews** in one platform.

Cooperative Gig Services Platform for household & community Services

SOFTWARE

- **Client Devices**-web/mobile (Android/IOS)for customer,workers and cooperative/admin.
- **React(Frontend)**-responsive web application with multilingual UI.
- **Python(Backend)**-FastAPI/Flask for rest API.
- **Machine Learning**-Scikit-learn for worker matching and demand forecasting.
- **Database**-PostgreSQL/Firebase
- **Maps and Geo-location**-Leaflet +OpenStreetMap.
- **Version control**-Git and Github.
- **Authentication and security**-
OTP-based verification,JWT,role-based access.



[coopserve](https://coopserve.org)

Feasibility

- Use our existing skills:**React,python/ML C++/DSA and UI/UX.**
- Open-sources technologies keep development **low cost and accessible.**
- Simple **React +python REST API** architecture is practical for our team.
- Location based matching can be implemented using **leaflet and OpenStreetMap.**
- Database/storage technology can be **finalized based on MVP requirements.**
- MVP can be developed and tested within the **SIH timeline.**

Challenges

- Ensuring **accurate worker matching** with limited initial data.
- Maintaining **reliable location and availability information.**
- Verifying workers **skills,identity and service quality.**
- Handling **real-time booking and status updates** efficiently.
- Protecting **user data** and preventing **fraudulent ratings or bookings.**
- Scaling storage,APIs and ML models as **the number of users grows.**

Strategies to overcome challenge

- **Finding the right worker-location** based matching using skills,availability,and service and service requirement.
- **Trust and safety issues**-aadhar id based verification,background verification and user rating.
- **Payment and fraud risk**-secure digital payments with transaction tracking and dispute resolution.
- **Data privacy**-secure authentication encrypted data and controlled access to personal information.
- **Emergency/safety concern**-SOS feature,emergency contacts and service tracking.
- **Scalibility**-modular platform architecture that can expand from local communities to multiple cities.

Impact on Key Stakeholders

Customers-

- Easy access to trusted and verified local workers.
- Convenient ,safe and affordable service booking .
- Saves time and effort.

Workers-

- More local jobs opportunities
- Better utilization of skill and times.
- Transparent earning and fair ratings.

Cooperatives-

- Centralised management of workers and services.
- Transparent record of booking and earnings.
- Data-driven insight for better resource allocation.

Community /society

- Reliable and quality household work.
- supports local employment and skill development.
- Build trust and social inclusion.



Economic and Strategic Benefits

Cost saving-

- Reduces dependency on third-party platforms.
- Low operational cost using open-source technologies

Economic growth

- Small services/cooperative fee ensure sustainability.
- Large markets potential in household and community services.

Scalability-

- Can start with one locality and scale to multiple cooperative and regions.
- Scope for future integrations (eg-digital payments,more service categories).

Long term values

- Promote local employment and skill development.
- Supports government's vision of a self-reliant and inclusive India .

RESEARCH AND REFERENCES

RESEARCH WORK

- **Studied existing platform**(urban company,TaskRabbit,BharatMitr)to understand service booking,worker management and user experience.
- **Researched Cooperative Models** and government initiatives to align with community-based and self-reliant approach.
- **Explored location-based service** matching using open-source tools like OpenStreetMap and Leaflet.
- **Analysed machine learning techniques** for recommendation and ranking based on skills,location and availability.
- **Reviewed literature on gig economy** ,local employmen and digital platforms to assess impact and scalability.
- **Identified key gaps** in existing solutions such as high cost,lack of cooperative involvement and limited focus on local workforce.

REFERENCES AND LINKS

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